

# Neil Ashton

Vice President, Computational Engineering, Mistral

neilashton.co.uk | contact@neilashton.co.uk | Google Scholar | ORCID | GitHub | LinkedIn

Computational engineering researcher and technical leader working at the intersection of scientific machine learning, high-fidelity simulation, foundation models, and agentic AI for verifiable and traceable engineering workflows. Experience spans research and product direction, open scientific datasets, international benchmarking programmes, high-performance computing, and industrial deployment.

## Experience

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| <b>Sep 2026–present</b>  | <b>Vice President, Computational Engineering</b><br>Mistral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Jan 2025–Sep 2026</b> | <b>Distinguished Engineer</b><br>NVIDIA   United Kingdom <ul style="list-style-type: none"><li>- Set research and product direction for accelerated computational engineering, AI physics, simulation, and engineering AI.</li><li>- Led work on foundation models, scientific machine learning, and agentic architectures, with emphasis on tool orchestration, physical verification, provenance, uncertainty, failure recovery, and human oversight.</li><li>- Worked across research, engineering, product, and external partners to translate advances into dependable computational-engineering systems.</li></ul> |
| <b>Feb 2020–Jan 2025</b> | <b>Worldwide Technical Lead for Computer-Aided Engineering</b><br>Amazon Web Services <ul style="list-style-type: none"><li>- Led worldwide technical strategy for CAE across high-performance computing, machine learning, cloud services, and partner technologies.</li><li>- Founded and led a cross-Amazon machine-learning-for-CAE initiative, connecting product development, research, and industrial adoption.</li><li>- Provided technical leadership for open, high-fidelity datasets including AhmedML, WindsorML, and DrivAerML.</li></ul>                                                                   |
| <b>Jul 2016–Jan 2020</b> | <b>Senior Researcher</b><br>Osney Thermofluids Institute, Department of Engineering Science, University of Oxford <ul style="list-style-type: none"><li>- Led vehicle-aerodynamics research spanning high-fidelity CFD, turbulence modelling, multiphysics, and high-performance computing.</li><li>- Supervised doctoral research and led collaborations with academic, government, and industrial partners.</li><li>- Founded Oxford University Racing and chaired the 10th International Conference on Computational Fluid Dynamics.</li></ul>                                                                        |
| <b>Jul 2017–Dec 2019</b> | <b>Consultant</b><br>Formula 1 Management <ul style="list-style-type: none"><li>- Provided CFD and HPC expertise to the motorsport team, co-authored the CFD/HPC sections of proposed FIA Sporting Regulations, reviewed Technical Regulations, and helped establish a cloud-based OpenFOAM process.</li></ul>                                                                                                                                                                                                                                                                                                           |
| <b>Jul 2016–Dec 2019</b> | <b>Consultant</b><br>British Cycling <ul style="list-style-type: none"><li>- Conducted the high-fidelity CFD programme for the Great Britain track bicycle developed for the Tokyo 2020 Olympic Games, including more than 1,000 transient simulations.</li></ul>                                                                                                                                                                                                                                                                                                                                                        |
| <b>Jan 2016–Jul 2017</b> | <b>Consultant</b><br>Williams Advanced Engineering <ul style="list-style-type: none"><li>- Established CFD and aerodynamic methods for Formula 3, Le Mans, and touring-car programmes.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Jan–Jul 2016</b>      | <b>Visiting Scholar</b><br>Advanced Supercomputing Division, NASA Ames Research Center <ul style="list-style-type: none"><li>- Conducted large-scale computational aerodynamics research for transonic and supersonic aircraft with the Applied Modeling and Simulation group.</li></ul>                                                                                                                                                                                                                                                                                                                                 |
| <b>2013–2015</b>         | <b>EPSRC Impact Acceleration Research Associate</b><br>University of Manchester <ul style="list-style-type: none"><li>- Developed turbulence models for automotive and aerospace applications and contributed to the EU FP7 Go4Hybrid programme.</li><li>- Undertook a one-month research visit to NASA Ames Research Center in February 2015.</li></ul>                                                                                                                                                                                                                                                                 |
| <b>2012–2013</b>         | <b>CFD Engineer</b><br>Lotus F1 Team <ul style="list-style-type: none"><li>- Developed computational-fluid-dynamics and high-performance-computing methods within the Formula 1 aerodynamics group.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                            |

## Education

2008–2013

### PhD, Aerospace Engineering

University of Manchester

Thesis: Development, Implementation and Testing of an Alternative DDES Formulation Based on Elliptic Relaxation. EPSRC funded.

2004–2008

### MEng (Hons), Aerospace Engineering – First Class

University of Manchester

## Professional standing

|                                                                      |            |
|----------------------------------------------------------------------|------------|
| Fellow, Institution of Mechanical Engineers                          | Since 2020 |
| Chartered Engineer, Institution of Mechanical Engineers              | Since 2013 |
| Senior Member, American Institute of Aeronautics and Astronautics    | Since 2014 |
| Member, NAFEMS CFD Working Group                                     | Since 2014 |
| Expert reviewer, European Commission                                 | Since 2017 |
| Member, IMechE Technical Strategy Board and Education Advisory Group | 2017–2021  |

## Research leadership and public scholarship

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|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019–present | <b>Automotive CFD Prediction Workshop (AutoCFD)</b><br>Founder and organiser<br>Founded the international series for open test cases, reproducible comparisons, and shared automotive-CFD research priorities. |
| Current      | <b>AIAA CFD High-Lift Prediction Workshop (HLPW)</b><br>Organising Committee and AI/ML Technical Focus Group Lead<br>Directs aerospace benchmarking and leads machine-learning data and evaluation activities. |
| Current      | <b>CAE ML Datasets</b><br>Research lead and collaborator<br>Leads the open AhmedML, WindsorML, DrivAerML, and HiLiftAeroML collection of more than 3,100 high-fidelity simulation samples.                     |
| 2024–present | <b>The Neil Ashton Podcast</b><br>Founder and host<br>Long-form conversations with research and engineering leaders in AI, scientific computing, simulation, and aerodynamics.                                 |

## Selected publications

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|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. 2026 | Neil Ashton et al. <a href="#">HiLiftAeroML: High-Fidelity Computational Fluid Dynamics Dataset for High-Lift Aircraft Aerodynamics</a> . <i>arXiv preprint</i> .                                                           |
| 2. 2025 | Neil Ashton, Johannes Brandstetter, and Siddhartha Mishra. <a href="#">Fluid Intelligence: A Forward Look on AI Foundation Models in Computational Fluid Dynamics</a> . <i>arXiv preprint</i> .                             |
| 3. 2024 | Neil Ashton et al. <a href="#">WindsorML: High-Fidelity Computational Fluid Dynamics Dataset for Automotive Aerodynamics</a> . <i>Advances in Neural Information Processing Systems</i> 37, 37823–37835.                    |
| 4. 2024 | Neil Ashton, Paul Batten, Andrew Cary, and Kevin Holst. <a href="#">Summary of the 4th High-Lift Prediction Workshop Hybrid RANS/LES Technology Focus Group</a> . <i>Journal of Aircraft</i> 61(1), 86–115.                 |
| 5. 2023 | Neil Ashton and William Van Noordt. <a href="#">Overview and Summary of the First Automotive CFD Prediction Workshop: DrivAer Model</a> . <i>SAE International Journal of Commercial Vehicles</i> 16(1), 61–85.             |
| 6. 2021 | Jamil Appa, Mike Turner, and Neil Ashton. <a href="#">Performance of CPU and GPU HPC Architectures for Off-Design Aircraft Simulations</a> . <i>AIAA SciTech</i> 2021.                                                      |
| 7. 2016 | Neil Ashton, Alastair West, Sylvain Lardeau, and Alistair Revell. <a href="#">Assessment of RANS and DES Methods for Realistic Automotive Models</a> . <i>Computers &amp; Fluids</i> 128, 1–15.                             |
| 8. 2013 | Neil Ashton, Alistair Revell, Robert Prosser, and Juan Uribe. <a href="#">Development of an Alternative Delayed Detached-Eddy Simulation Formulation Based on Elliptic Relaxation</a> . <i>AIAA Journal</i> 51(2), 513–519. |

## Selected invited talks

|          |                   |                                                                                                                                                                          |
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| Jul 2026 | Sponsored keynote | <b>SIGGRAPH 2026</b><br>NVIDIA - Next Era of Graphics - Neural Rendering, World Models, and Simulation                                                                   |
| May 2026 | Invited seminar   | <b>University of Oxford Mathematical Institute</b><br>Towards a Foundation Model for Computational Engineering: Opportunities, Challenges, and Novel Scaling Laws        |
| Apr 2026 | Invited seminar   | <b>NASA Ames Seminar Series</b><br>Perspectives on Foundation Models and Agentic Architectures for Computational Fluid Dynamics                                          |
| May 2025 | Invited talk      | <b>NASA High-Fidelity CFD Workshop</b><br>Enabling Industrially Relevant High-Fidelity CFD and AI Surrogate Models for External Aerodynamics                             |
| Apr 2023 | Invited keynote   | <b>22nd International Computational Fluids Conference</b><br>The Role of Cloud Computing, Machine Learning, and HPC in the Advancement of High-Fidelity CFD for Industry |
| Mar 2023 | Invited seminar   | <b>MIT ACDL Seminar Series</b><br>The Role of Cloud Computing, Machine Learning, and HPC in the Advancement of High-Fidelity CFD for Industry                            |